DESCENT DATA FOR A FAMILY EQUAL DESCENT DATA FOR THE COPRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Allowed standard prerequisites (editorial).

- Descent data relative to a morphism and to a family. chapter definitions definition-descent-datum and definition-descent-datum-for-family-of-morphisms. The two categories being equated.

Lemma 0.1. *Let S be a scheme. Let $\{X_i \to X\}_{i\in I}$ be a family of morphisms of algebraic spaces over S with fixed target X. Set $Y = \coprod_{i\in I} X_i$. There is a canonical equivalence of categories*

$$\begin{array}{c}\text{category of descent data}\\ \text{relative to the family } \{X_i \to X\}_{i\in I}\end{array} \longrightarrow \begin{array}{c}\text{category of descent data}\\ \text{relative to } Y/X\end{array}$$

which maps (V_i, φ_{ij}) to (V, φ) with $V = \coprod_{i\in I} V_i$ and $\varphi = \coprod \varphi_{ij}$.

Proof. Observe that $Y \times_X Y = \coprod_{ij} X_i \times_X X_j$ and similarly for higher fibre products. Giving a morphism $V \to Y$ is exactly the same as giving a family $V_i \to X_i$. And giving a descent datum φ is exactly the same as giving a family φ_{ij}. □

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